

CHEM 515 Practice Session 1: Building molecules using IQmol; Hartree-Fock energy calculations; Basis set convergence

Name: _____

In this practice, you are expected to learn how to use IQmol and Q-Chem (the quantum chemistry package on the back end) to build molecules and run basic quantum chemistry calculations using Hartree-Fock theory. You will also learn how to look into the results of Q-Chem calculations using IQmol.

You will need to **fill out the tables** and **complete the follow-up questions** on this handout. There are also additional tasks such as **generating graphics or other files** using the IQmol software. Please refer to the **Assignment page on Canvas** for detailed submission guidelines. The grading will be based on attendance (2 pts) and submission of your work (8 pts).

Exercise 1: Hartree-Fock calculation for a hand-crafted pyridine

In this exercise, you will build a pyridine molecule from scratch using IQmol and calculate its energy using HF/6-31G(d); you will then compare it with (i) the HF energy of the built-in structure of pyridine in IQmol, and (ii) the optimized structure of pyridine at the HF/6-31G(d) level. First, let's build a pyridine molecule with the following steps:

- Open IQmol. You should already be in the “Build” mode. Now just click anywhere on the Viewer (the part with a blue background), you should see a carbon atom appear (shown as a grey sphere).
- Click on the C atom and drag your mouse towards any random direction. You should see another C atom appear which forms a single bond with the original C atom. Keep doing that until 6 carbon atoms are added to form a hexagon (don't worry about how distorted it looks).
- Click “Build Element” on the toolbar at the top (the button that shows the element symbol), and select nitrogen (“N”). Click on any one of the 6 carbon atoms you drew to replace it with nitrogen.
- To ensure that the correct number of H atoms are added, we need to add 3 *alternating double bonds* to the 6-member ring. To create a double bond between two atoms that are already connected by a single bond, you just need to drag between them one more time. After adding 3 double bonds (two C=C and one C=N), click the “Add Hydrogens” button at the top. Each C atom should then get one H added.
- To provide a more reasonable starting structure for your calculations, IQmol can optimize the hand-crafted structures using a built-in force field. To do that, simply click the “Minimize Energy” (“↓E”) button on the toolbar.

Completing the above procedure should give you a decent-looking structure of the pyridine molecule. Now you are ready to set up your first HF calculation:

- Click “Calculation → Q-Chem Setup”, leave the job type (the drop-down menu of “Calculate”) and “Method” *unchanged* since a HF energy calculation is exactly what we need
- Change the “Basis” to “6-31G*” (which is equivalent to 6-31G(d)) and set “Convergence” to 8 (corresponding to keyword SCF_CONVERGENCE = 8)
- The job control panel should look like in Fig. 1. Now click “Submit” (make sure that the name of server is “Q-Chem”). You will be asked to give a name to your job, for which you can put in something like “pyridine_handdraw”. The job will then be executed by a remote server hosted at Q-Chem.

Figure 1: Job setup for pyridine HF/6-31G(d) calculation

Once the job finishes, IQmol will ask you to copy it back from the remote server. It is recommended that you create a specific parent directory on your computer for all the jobs in this practice (e.g. “**practice_1**”) and then create individual sub-directories for each of the jobs so that files won’t get accidentally overwritten. You are now asked to **report the obtained HF energy in Table 1**, under “pyridine (manually built)”. To get that value, you can simply expand the job that just finished (“**pyridine_handdraw**”) under the Model View (the column on the left), and then further expand “Geometries”. The number shown there corresponds to the HF/6-31G(d) energy of the pyridine structure you built.

You now need to complete a second HF calculation, using the structure of pyridine available in IQmol’s library of molecular structures:

- Click “New Molecule” (the first button on the toolbar). You should see the Viewer get cleaned up and a new item “Untitled” appear under Model View
- Click “Add Fragment” (the button on the left of “Add Hydrogens”), then select “Molecules” → “Heteroaromatics”, under which you can find “pyridine”. Select that and then click the “Select” button to close the window.
- Now you are under the “Build” mode. Click at a random place on the Viewer, you should see a pyridine molecule show up. Use the same settings as above to run a HF/6-31G(d) energy calculation. **Report the obtained energy under “pyridine (built-in)” in Table 1.**

Finally, we are going to evaluate the energy of the *optimized* structure of pyridine at the HF/6-31G(d) level. You can just start from the structure you used for your second calculation. **The only change you need to make is to set the job type (the menu next to “Calculate”) to “Geometry”**. While we haven’t covered geometry optimization in lectures yet, you can understand it as the procedure to find the structure of *minimum energy*. Submit the geometry optimization job. Once finished, under “Geometries” you can find multiple energy values in a descending order, which shows the work done by the optimizer (you can double-click “Geometries” under model view to see a plot of energies in each optimization cycle). **Report the optimized energy (the last number) under “pyridine (optimized)” in Table 1.**

Table 1: Results of HF/6-31G(d) calculations for various structures of pyridine (1.5 pt)

Structure	pyridine (manually built)	pyridine (built-in)	pyridine (optimized)
Energy (a.u.)			

Note: the energies should be reported with **6 decimals**.

Follow-up questions & additional tasks:

1. Based on the energy metric, is your hand-crafted structure of pyridine better or worse than the built-in structure? **Provide a quantitative assessment by calculating the energy difference between them (in kcal/mol). (0.25 pt)** (Note: 1 a.u. = 627.5095 kcal/mol)
2. Using the calculation for hand-crafted pyridine, visualize the HOMO and LUMO as 3D contour plots **(0.25 pt)**. Steps to follow:
 - Select your first finished calculation under “Model View” (make sure the other jobs are *unselected*), then double-click “Canonical Orbitals” under “Surfaces”. A window will pop up.
 - Set the range of orbitals to visualize from “Alpha 21 (HOMO)” to “Alpha 22 (LUMO)” (α and β orbitals are the same in this RHF calculation), and then click “Calculate”
 - The two orbitals will show up under “Canonical Orbitals” under Model View. Check them out by selecting one of them at a time.
 - It is recommended to turn off the blue background when we save figures using IQmol. To do that, click “Global” under Model View, and unselect “Background”. You can turn the blue background back on for later tasks if you prefer.
 - To save these 3D contour plots, select “File → Save Picture”, and then you can save the images in the “.png” format (recommended). Save the contour plots of HOMO and LUMO separately as “pyridine_homo.png” and “pyridine_lumo.png”. You should make sure that the two orbitals are not shown in the same image. These two images need to be submitted.
3. Based on the calculation using the hand-crafted structure, calculate the HOMO-LUMO energy gap. **Report the result in eV. (Hint:** the orbital energies in atomic unit (Eh) can be obtained either using IQmol (by clicking on the orbital energy levels in the MO diagram under the “Canonical Orbitals” window) or directly read from the output file. (1 Eh = 27.2114 eV) **(0.25 pt)**

Exercise 2: Basis set convergence of Hartree-Fock energy for H_3O^+

In this exercise, we are going to investigate how the HF energy for the H_3O^+ ion changes as the size of the basis set increases.

- Open the XYZ file **hydronium.xyz** (which can be downloaded from Canvas) using IQmol (click **File** → **Open** and then navigate to the directory where the .xyz file is stored). You may find that one of the O–H bond is missing, which doesn’t need to be worried about. When you press “**Ctrl** + **Y**” (or “command + Y” on Mac), IQmol should tell you that the structure is of C_{3v} symmetry.
- Set up a HF energy calculation with basis set “cc-pVDZ”. Note that the charge of this system needs to be set to “1”. In addition, the “Convergence” parameter under “SCF Control” should be **set to 8 throughout this exercise**.
- Once the job is finished, report the obtained **HF energy**, the **total number of basis functions**, and the **time duration** (“wall time”) of the job in **Table 2**. The energy can be found under “Geometries”, while the latter two numbers will need to be retrieved from the Q-Chem output file: under the current job in Model View, expand “Files” and double-click the “.out” file. A window will pop up. You can simply search for keywords “basis functions” to find out the # of basis functions; and scrolling down to the bottom you will see a line starting with “Total job time”, from which you can find the wall time.

- Repeat the above with cc-pVTZ, cc-pVQZ, and cc-pV5Z basis sets and **complete Table 2**. Note: (i) cc-pV5Z is not available under the “Basis” dropdown menu so you need to manually edit the generated input file (simply by typing); (ii) the calculation using cc-pV5Z will take a few minutes so please be patient.

Table 2: Hartree-Fock energies for H_3O^+ using different basis sets (**1.5 pts**)

Basis set	Energy (a.u.)	# of basis functions	Timing (s)
cc-pVDZ			
cc-pVTZ			
cc-pVQZ			
cc-pV5Z			

Follow-up questions & additional tasks:

1. Write out the shell structures ($ns(n-1)p(n-2)d\dots$) of the cc-pV5Z basis set for oxygen and hydrogen. You may want to review the discussion of correlation-consistent basis sets in Lec. 6. (**0.25 pt**)
2. Based on the HF energies you obtained using different basis sets, do you expect that the energy will eventually converge? Why or why not? (**0.25 pt**)
3. In our last lecture, we introduced that the computational cost of a HF calculation increases with the number of basis functions as $\mathcal{O}(n^4)$. Comparing the # of basis functions and timings for the cc-pVTZ and cc-pVQZ calculations, is the increase in computational cost (measured by the wall time of the job) steeper or less steep than the $\mathcal{O}(n^4)$ scaling? What about the comparison between cc-pVQZ and cc-pV5Z? (**0.5 pt**)

Exercise 3: Ethane’s rotational barrier and dissociation energy

In this exercise, you will calculate the **rotational barrier and the C–C dissociation energy of ethane** at the HF/cc-pVDZ level of theory. These two energetic quantities are defined as the follows:

$$\Delta E_b = E(\text{eclipsed}) - E(\text{staggered}) \quad (1)$$

$$\Delta E_d = 2E(\text{CH}_3) - E(\text{C}_2\text{H}_6) \quad (2)$$

Let’s first obtain the energetically optimized structure of the *staggered* conformation with the following steps:

- Start with “New Molecule”, then load a staggered ethane structure into IQmol which can be easily done by selecting “Add Fragment → Molecules → Alkanes → Ethane”. Symmetrize the molecule by “Ctrl + Y” (“Command + Y” on Mac). The symmetry group identified by IQmol should be D_{3d} .
- Submit a geometry optimization job (“Calculate → Geometry”) using HF/cc-pVDZ. Once the job finishes, report (i) **the optimized C–C bond length** and (ii) **the energy of the optimized structure** in **Table 3**. To obtain the bond length, you should first enter the “Select” mode by clicking the “magic wand” next to the red “X” on the toolbar. You can then click on the two carbon atoms: the bond length will show on the bottom left corner of the Viewer.
- Start over by clicking “New Molecule” and repeat the first step (load the ethane structure from IQmol’s built-in library). We are now going to build the *eclipsed* structure from that. First, hit the key “T” on the keyboard, which will let IQmol show the index of each atom in the XYZ structure. Then under the **Select mode**, click atoms “1–2–5–8” (note: you must follow this order) and then select “Build → Set Geometric Constraint”. You will see a small window pop up which indicates that the torsion (dihedral) between atoms 1–2–5–8 is 180° . You can now change this number to 0° and click “Apply”. This change will generate an *eclipsed* conformation. Symmetrize the structure using the same shortcut. The symmetry group of the resulting structure should be D_{3h} .
- Submit a geometry optimization job for the eclipsed conformation with the same settings. Report the equilibrium energy and C–C bond length in **Table 3**.

Table 3: Staggered and eclipsed ethane calculated at the HF/cc-pVDZ level (**1 pt**)

	staggered	eclipsed
C–C bond length (Å)		
Energy (a.u.)		

Based on the results in Table 3, calculate the rotational barrier (in kcal/mol) of ethane using Eq. (1). (**0.25 pt**)

To evaluate the dissociation energy, we further need to obtain the optimized structure for the methyl radical $\cdot\text{CH}_3$. Starting with an ethane structure, you can select all atoms in one of the methyl groups and then click “Delete Selection” (the icon with a red cross). You can then submit a geometry optimization job for the remaining part (the $\cdot\text{CH}_3$ radical) with the same settings, except that now the multiplicity is 2 (open-shell). **Report the energy of the optimized $\cdot\text{CH}_3$ below (0.25 pt):**

$$E(\cdot\text{CH}_3) =$$

Then, using Eq. (2) and the energy of the more stable *staggered* conformation, **calculate the bond dissociation energy** (in kcal/mol). Show your work below (**0.25 pt**):

Follow-up questions & additional tasks:

1. Save optimized ethane structures in the staggered and eclipsed conformations as XYZ coordinates. To do this, you need to select each of the geometry optimization jobs under Model View, and then click “File → Save as”, under which the default file type is .xyz. Name the two XYZ files as “staggered.xyz” and “eclipsed.xyz”, which need to be submitted. **(0.25 pt)**
2. The experimental value for the rotation barrier of ethane is 2.875 kcal/mol. What are the *signed* absolute and relative (%) errors of the barrier predicted by HF/cc-pVDZ? **Show your work below (0.25 pt):**
3. The average bond energy for C–C is **347 kJ/mol** (1 kcal = 4.184 kJ). Compared to that, does HF/cc-pVDZ over- or underestimate the C–C bond energy? Answer this question by calculating the signed absolute and relative errors. **Show your work below (0.25 pt):**
4. Based on the discussion of the main source of errors of HF theory in class, explain why the *absolute error* for ethane’s rotational barrier is much smaller than that for the dissociation energy of ethane’s C–C bond. **(0.25 pt)**
5. Using the staggered and eclipsed structures of ethane optimized at the HF/cc-pVDZ level, recalculate the rotation barrier using the cc-pVTZ basis. Complete Table 4 below. Is the agreement with experimental value improved? **(0.5 pt)** (**Note:** you should not reoptimize the structures; just recompute the energies. The method employed here can be denoted as HF/cc-pVTZ // HF/cc-pVDZ)

Table 4: Recalculation of the rotational barrier of ethane using HF/cc-pVTZ

Staggered (a.u.)	Eclipsed (a.u.)	ΔE_b (kcal/mol)